

Repartitie normală

$$X \sim \mathcal{N}(m, \sigma^2) \quad (\text{în } \mathbb{R} \text{ din } \mathcal{N}(2, 3))$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-m)^2}{2\sigma^2}}$$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \rightarrow \text{nu se poate calc. explicit}$$

OBS. Pt. $m=0, \sigma=1$ $X \sim \mathcal{N}(0, 1)$ normală standard

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}}$$

$$\Phi(x) = \int_{-\infty}^x f(t) dt \rightarrow \text{valori tabelate}$$

$\hookrightarrow$ Gauss-Laplace $\Phi(-x) = 1 - \Phi(x)$ ♥

Procedul de standardizare

$$X \sim \mathcal{N}(m, \sigma^2)$$

$$Z = \frac{X-m}{\sigma} \sim \mathcal{N}(0, 1)$$

$$F_Z(x) = \mathbb{P}(Z \leq x) = \mathbb{P}\left(\frac{X-m}{\sigma} \leq x\right) = \mathbb{P}(X \leq \sigma \cdot x + m) = F_X(\sigma \cdot x + m)$$

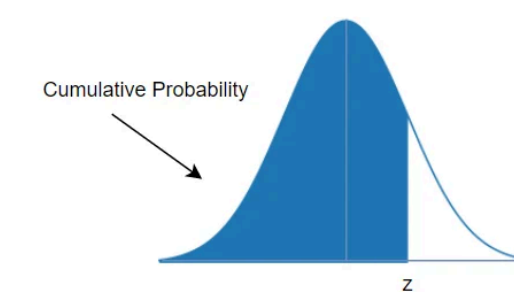
$$\stackrel{\sigma x+m}{=} \int_{-\infty}^{\sigma x+m} f_X(t) dt = \int_{-\infty}^{\sigma x+m} \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(t-m)^2}{2\sigma^2}} dt$$

S.V. $\frac{t-m}{\sigma} = y \Rightarrow \boxed{t = \sigma y + m} \rightarrow dt = \sigma dy$

$$t \rightarrow -\infty \Rightarrow y \rightarrow -\infty$$

$$t = \sigma x + m \Rightarrow y = x$$

$$F_Z(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(\sigma y + m - m)^2}{2\sigma^2}} \cdot \sigma dy = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{y^2}{2}} dy = \Phi(x)$$



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7938	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9405	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9685	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9725	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9825	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9959	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

0.02 coloana

$$\Phi(1.52) = 0.9357$$

linia

$$\Phi(1.525) \approx \frac{0.9357 + 0.9370}{2}$$

$$\mathbb{P}((X > 290) \cup (X < 240)) = \mathbb{P}(X > 290) + \mathbb{P}(X < 240)$$

INC.

$$\mathbb{P}(X > 290) = 1 - \underbrace{\mathbb{P}(X \leq 290)}_{F_X(290)} = 1 - F_X(290)$$

$$\mathbb{P}(X < 240) = \mathbb{P}(X \leq 240) = F_X(240)$$

Aplic procedul de standardizare

$$\mathbb{P}(X > 290) = \mathbb{P}\left(\frac{X-270}{10} > \frac{290-270}{10}\right) = \mathbb{P}(Z > 2) = 1 - \mathbb{P}(Z \leq 2) = 1 - \Phi(2)$$

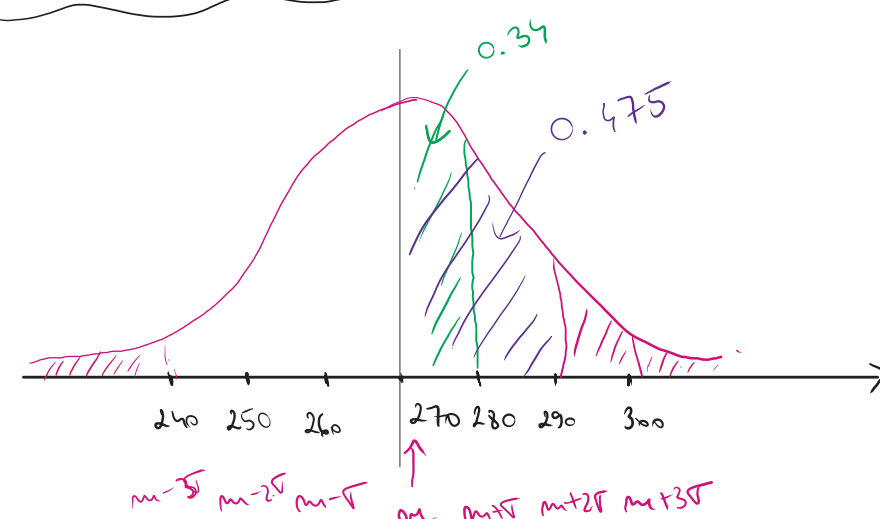
$$\mathbb{P}(X \leq 240) = \mathbb{P}\left(\frac{X-270}{10} \leq \frac{240-270}{10}\right) = \mathbb{P}(Z \leq -3) = \Phi(-3) = 1 - \Phi(3)$$

$$\mathbb{P}((X > 290) \cup (X \leq 240)) = 1 - \Phi(2) + 1 - \Phi(3) = 2 - \Phi(2) - \Phi(3) = 2 - 0.9772 - 0.9987 \approx 0.0241$$

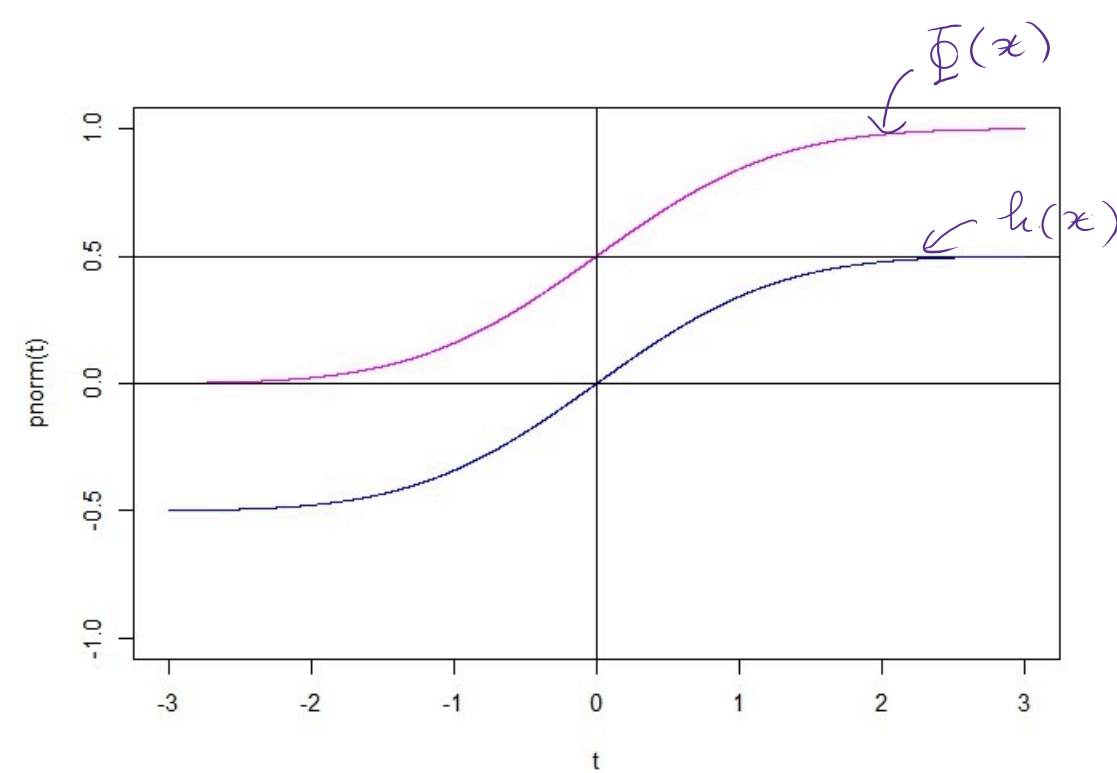
"Sigel poate fi fatal lui Bebe"

Concluzie: Prob. ca Sigel sa poata fi fatal lui Bebe este $\approx 2,41\%$.

Regula celor 3 σ



Dem. pt. ♥: $\Phi(-x) = 1 - \Phi(x)$



$$\Phi(x) = h(x) + \frac{1}{2}$$

$$h \rightarrow \text{impare} \Rightarrow h(-x) = -h(x)$$

$$\Phi(-x) = h(-x) + \frac{1}{2} = -h(x) + \frac{1}{2} = -h(x) - \frac{1}{2} + 1 = 1 - \underbrace{\left(h(x) + \frac{1}{2}\right)}_{\Phi(x)} = 1 - \Phi(x)$$